

5(a)	p.d. across R = 1.3 V Let R_R be the resistance across R. Using potential divider rule, $\frac{R_R}{0.75 + R_R + 5.5} \times 4.0 = 1.3$ $R_R = 3.0 \, \Omega$ Hence, $\text{p.d. across AB} = \frac{5.5}{0.75 + 3.0 + 5.5} \times 4.0$ $= 2.38$ $= 2.4 \, \text{V (2 s.f.)}$	
5(b)(i)	$E = \frac{1.5 - 0.56}{1.5} \times 2.4$ $= 1.5 \, \text{V}$	
5(b)(ii)	The potential at C is higher than that at B. As C is shifted closer to A, the potential at C increases, thus increasing the potential difference between BC. Since the potential between BC will become larger than the terminal p.d. of cell Q, current will now flow through cell Q from C to B. OR This will lead to a potential difference across the internal resistance in cell Q, such that current will now flow through Q from C to B.	

5

No.	Solution	Remarks
06/11/20	5	